

Design solar system Installation

1. **Solar system**: Is a set of components that work together to harness solar energy and convert it into usable electricity for residential, commercial or industrial applications.
 - It typically includes solar panels, an inverter, mounting hardware and sometimes a battery bank for off-grid systems.
2. **Solar panels**: Also known as photovoltaic (PV) panels.
 - These are devices that convert sunlight into electricity through the photovoltaic effect.
 - They are made up of photovoltaic cells that generate direct current (DC) electricity when exposed to direct sunlight.
3. **Mono crystalline Panels**: These are panels made up from a ~~single~~ single crystal of silicon and have a higher efficiency rate.
4. **Poly crystalline Panels**: These are panels made up from multiple small silicon crystals and have a lower efficiency rate.
5. **Inverter**: Is a component of a solar system that converts the direct current (DC) electricity generated by solar panel into alternating current (AC) electricity that can be used in homes or fed into the electrical grid.
6. **Battery Bank**: Is a set of rechargeable batteries used in off-grid solar systems to store excess energy generated during the day for nighttime usage when there is less sunlight available.
7. **Grid-Tied system**: Is a type of solar system installation that connects to the electrical grid, allowing homeowners to sell excess energy back to the utility company through net metering programs.
8. **Net Metering**: Is an arrangement where homeowners or businesses can generate their own electricity using renewable sources like solar power and export any excess energy back to the electrical grid in exchange for credit on their utility bills.
9. **Ground Fault Circuit Interrupters (GFCIs)**: Are safety devices installed in electrical systems that detect and interrupt ground faults, providing protection against electric shocks and reducing the risk of electrocution.
10. **Surge Protection Device (SPDs)**: Are components installed in solar systems to protect sensitive electronic equipment from voltage spikes and surges caused by

lightning strikes or other power fluctuations.

Size and rating of Solar Panel

- The size and rating of the solar panel determines the amount of electricity it can generate per hour when exposed to sunlight.
- Larger panels with higher wattage ratings can produce more electricity but may require more space on the roof or ground.

To calculate size of solar panel:

- Assess energy consumption
- Determine sunlight hours
- Calculate the required wattage
- Consider roof space and orientation
- Consult a professional

Factor Affecting Solar Panel Efficiency

Efficiency refers to how well a solar panel converts sunlight into electricity.

Factors

- Temperature
- Angle of Incidence
- Shading
- Dirt accumulation

Solar Panel Mounting Positioning

- Roof orientation
- Tilt Angle
- Azimuth Angle - refers to the compass direction in which the solar panels are facing.
→ panels should face south in northern hemisphere and north in south.
- Shading
- Ground Mounting
- Roof Mounting
- Professional Installation

Components of Solar System

- **Solar panel** : Convert sunlight into electricity
- **Inverter** : Converts DC electricity from solar panels to AC electricity for use in homes or feeding into the grid
- **Battery Bank** : Stores excess energy generated during the day for nighttime usage in off-grid systems.
- **Grid-Tied system** : Connects to the electrical grid, allowing homeowners to sell excess energy back through net metering programs.
- **Net metering** : Allow homeowners to generate their own electricity and export excess energy back to the grid in exchange for credit on utility bills.
- **Ground Fault Circuit Interrupters (GFCIs)** : Protect against electric shock by detecting and interrupting ground faults.
- **Surge Protection Devices (SPDs)** : Protect sensitive electronic equipment from voltage spikes and surges caused by lightning strikes or power fluctuations.

Solar Charger Controller.

A device that regulates the flow of electricity from solar panels to batteries or other loads, ensuring efficient energy storage and use.

- ⇒ The controller optimizes energy harvesting by constantly adjusting the voltage and current of the solar panels to their maximum power point.
- ⇒ It maintains proper battery charge levels by preventing overcharging and undercharging, prolonging battery lifespan.
- ⇒ Also regulate the voltage and current supplied to electrical loads, ensuring stable power delivery.
- ⇒ Solar controller often include protection features such as overcharge protection, over discharge protection, short circuit protection and reverse polarity protection.
- ⇒ High efficiency controllers maximizes the transfer of energy from the solar panels to the batteries or loads, minimizing energy ~~loss~~ waste.

SIZE AND RATING OF SOLAR PANE CALCULATIONS

Step 1. Home electrical load calculation = 3240 kWh

2: Solar panel selection

3: Battery selection

4: Inverter

5: Solar charge controller.

Load Name	Power of load (W)	No of Load	Total power W	Operating hours (h)	Total power kWh
Led light	10	4	40	6	240
Tube light	18	4	72	5	360
Ceiling Fan	40	3	120	8	960
TV	50	1	50	8	400
Refrigerator	160	1	160	8	1280
			<u>442W</u>		<u>3240</u>

② Solar panel selection

Total power required per Day = 442W

Total energy = 3240 kWh

$$\text{Power required from Solar Panel (PW) W} = \frac{\text{Total Energy required from solar PV (Wh)}}{\text{Effective sunshine hours} \times \text{System efficiency } \eta_s}$$

$$= \frac{3240 \text{ kWh}}{7.5 \times 0.73} = 591.73 \text{ W}$$

effective sunshine hours in kwh = 7.5

$$\text{System efficiency } (\eta_s) = \eta_b \times \eta_i \times \eta_{oh}$$

η_b Battery efficiency (Assume 90%)

η_i Inverter (Assume 95%)

η_{oh} other components (charge controller, cable, etc.) (Assume 85%)

$$\eta_s = 0.73$$

$$\text{Power required from Solar Panel} = \frac{3240 \text{ kWh}}{7.5 \times 0.73} = 591.73 \text{ W} \approx \underline{\underline{600 \text{ Watts}}}$$

$$\text{Number of panels required} = \frac{\text{Power required for solar panel (wh)}}{\text{Output power of solar panel (wh)}}$$

$$= \frac{800 \text{ W}}{200 \text{ Each panel}} = \underline{\underline{4 \text{ panels}}}$$

③ . Battery selection

$$\text{Total power required Per Day} = 442 \text{ W}$$

$$\text{T.E} \quad \text{''} \quad \text{''} \quad \text{''} \quad \text{''} = 3240 \text{ Wh}$$

$$\begin{aligned} \text{Total required Battery Capacity (AH)} &= \frac{\text{Total Back up Energy required Per day}}{\text{Battery voltage} \times \text{Depth of Discharge} \times \text{Battery efficiency}} \\ &= \frac{3240 \text{ Wh}}{12 \times 0.85 \times 0.85} = \underline{\underline{400 \text{ AH}}} \end{aligned}$$

Number of Battery required

$$\Rightarrow \frac{\text{Required Battery Capacity (AH)} \times \text{Backup day}}{\text{Required AH of 12V Battery}}$$

We are take 12V 200AH

$$= \frac{400 \text{ AH} \times 1}{200 \text{ AH}} = 2 \text{ Battery}$$

$$200 \text{ AH} \times 2 \text{ Battery} = 400 \text{ AH}$$

④ Inverter selection

$$\text{Total power required per day} = 442 \text{ W}$$

$$\text{Required Inverter power} = \frac{\text{T.P} \times \text{Safety factor}}{\eta \text{ (Inverter efficiency 95)}}$$

$$\text{Safety factor} = 1.25$$

$$\Rightarrow \frac{442 \text{ W} \times 1.25}{0.95} = 582 \text{ Watts}$$

$$\approx 600 \text{ Watts}$$

⑤ Charger Controller:

$$\text{Charge Controller current} = \frac{\text{Total solar panel power w}}{\text{Battery Bank voltage (v)}}$$

$$= \frac{800 \text{ w}}{12 \text{ v}} = \underline{\underline{67 \text{ A}}}$$